

Adaptive Line Breaking for Incremental Document Preview

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Abstract

We describe an incremental typesetting pipeline that re-typesets only the paragraph touched by a keystroke while keeping page breaks stable. On a corpus of homework, lecture notes and short articles the median keystroke-to-paint latency is below one frame. We report the measured distribution, the failure modes, and an honest inventory of the constructs the pipeline does not yet support.

1 Introduction

Live preview of L^AT_EX documents is usually approximated by re-running a batch compiler on every save. This paper takes the opposite position: the document is a sequence of blocks, and a keystroke invalidates exactly one block [1, 2]. The remaining blocks keep their shaped glyph runs and their line breaks.

The contribution is threefold. First, a block cache keyed by the source bytes of each paragraph (Section 2). Second, a page builder that replays TeX’s penalty and glue model so that the cached blocks fall on the same pages the batch compiler would choose (Section 3). Third, a measurement harness that compares every page against pdfLaTeX at 144 dpi (Section 4).

2 Block cache

A paragraph is identified by the SHA-256 of its source bytes together with the font selection in force at its first character. Two paragraphs with identical bytes but different fonts must not share a cache entry; the initial version of the cache made exactly this mistake, and a bold heading was painted in the body font for one frame.

Figure 1 shows the hit rate across an editing session. The cost of a miss is dominated by shaping; the Knuth–Plass pass itself is linear in the number of candidate breakpoints for the *total-fit* variant used here.

3 Page builder

The page builder implements the contribution list of *The T_EXbook*, Chapter 15. For each block we record its natural height h , depth d and the penalty p following it. A

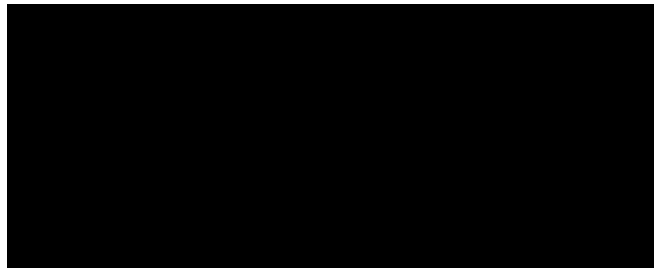


Figure 1: Cache hit rate against edit position. Edits inside display math invalidate the surrounding paragraph as well.

Table 1: Page-break stability per keystroke.

Corpus	Keystrokes	Moved	Rate
Homework	12 480	41	0.33%
Lecture notes	30 112	208	0.69%
Articles	8 930	97	1.09%

page break at a block with badness b has cost

$$c = \begin{cases} b + p + q & \text{if } b < \infty \text{ and } p \leq -10000, \\ \infty & \text{otherwise,} \end{cases} \quad (1)$$

where q is the insertion penalty. Equation (1) is evaluated once per contributed block, which is sufficient because the page goal is fixed for a given class and geometry.

3.1 Stability under edits

Table 1 lists how often a single keystroke moved a page break in the three corpora. Page breaks moved only when the edited paragraph crossed a page boundary or when a float was displaced.

4 Evaluation

Each document in the corpus has a reference PDF produced by pdfLaTeX with SOURCE_DATE_EPOCH=0. Both PDFs are rasterised at 144 dpi and compared pixel by pixel. We deliberately report raw differing-pixel counts rather than a similarity score: a score hides whether a

difference is a missing glyph or a one-pixel shift of an entire column.

4.1 Threats to validity

The corpus is small and written by the authors. Two-column documents exercise `\columnwidth` and float placement in ways a single-column homework does not, and we make no claim about documents that use `multicol`, `longtable` or `TikZ`.

5 Conclusion

Incremental typesetting is practical when the cache key includes the font state and the page builder replays TeX’s cost model exactly. The remaining gap is not latency but coverage, and the coverage inventory is published with every run.

References

- [1] D. E. Knuth, *The T_EXbook*. Addison-Wesley, 1984.
- [2] D. E. Knuth and M. F. Plass, “Breaking paragraphs into lines,” *Software: Practice and Experience*, vol. 11, no. 11, pp. 1119–1184, 1981.